Burkina Faso: Debt Sustainability Analysis

1. This appendix assesses the sustainability of Burkina Faso's external public debt using the Debt Sustainability Analysis (DSA) template for low-income countries. The DSA was conducted jointly by Fund and Bank staff and included a loan-by-loan reconciliation of the multilateral debt data. The macroeconomic scenario underlying this DSA is summarized in Box 2 of the appendix. Based on the analysis of the baseline scenario and several alternative scenarios, the staffs have come to the view that the country faces a moderate risk of debt distress.

Burkina Faso's debt situation at end-2004

2. The focus of the DSA is on Burkina Faso's external debt situation. While significant in the past, domestic debt has declined rapidly in recent years in part because of requirements by the West African Economic and Monetary Union – to which Burkina Faso belongs – to phase out indebtedness to the common central bank. In 2005, domestic debt represents only 5 percent of GDP, and it is projected to decline further over the medium term.

3. Burkina Faso's external debt stock net of HIPC debt relief stood at USD 1.75 billion at end-2004.[13] Multilateral creditors accounted for 72.6 percent of this debt, including IDA (49.2 percent), the African Development Bank (18.1 percent) and the IMF (5.3 percent). Most of the bilateral debt is held by non-Paris Club members, some of whom have not yet started providing their share of debt relief.

4. In April 2002, Burkina Faso reached the completion point under the Enhanced HIPC Initiative. At the time, it was judged that exogenous factors outside of the government's control (such as falling cotton prices) had contributed to a significant deterioration of Burkina Faso's debt outlook and the country became the first HIPC to obtain an additional topping-up under the HIPC Initiative. At end-2004, Burkina Faso's debt situation looked more optimistic than had been projected at the completion point (Box 1). Despite adverse exchange rate movements and higher than anticipated new borrowing, at end-2004, the country had an NPV of debt-to-exports ratio that was lower than the one projected at the completion point.

5. The latest World Bank's Country Policy and Institutional Assessment (CPIA) ranks Burkina Faso as a 'strong performer'. Indicative debt-burden thresholds for countries in this category are an NPV of debt-to-exports of 200 percent; to GDP of 50 percent; and to

[13]The debt stock net of obtained HIPC debt service relief is used to measure the existing debt burden of the country. The actual debt outstanding reported by the country and the creditors was USD2.1 billion at end-2004. The two numbers differ because HIPC debt relief is provided only in part as a reduction of the debt stock. Most creditors provide relief in the form of grants to help cover debt service obligations for a limited period or in the form of a debt rescheduling.

revenues of 300 percent; and debt service ratios to exports and to revenues of 25 and 35 percent, respectively.

Box 1. Burkina Faso: Decomposition of the Difference Between the End-2004 NPV of Debt-to-Exports Ratio Under the New Debt Sustainability Framework (DSF) and the Estimate at the Completion Point (CP)
(in percentage points)

		Changes due to methodology 2/	Changes due to economic variables
NPV of debt-to-exports ratio for end-2004 projected at time of completion point 1/	192.52		
(i) Parameters used to calculate NPV of debt	24.95		
Exchange Rate	26.56	33.36	(6.80)
Discount rate	(1.61)	(35.93)	34.32
(ii) New borrowing	7.43		
Volume	5.85		
Concessionality	1.58		
(iii) Exports	(71.54)	(18.41)	(53.13)
(iv) Residual 2/	48.56		
NPV of debt-to-exports ratio at end-2004 as of DSF 3/	183.11		

1/ The difference between the NPV of debt-to-exports ratio as of the DSF and the NPV of debt-to-exports ratio as projected at the CP equals the sum of lines (i), (ii), (iii) and (iv).
2/ Impact of using the WEO exchange rate projections and a 5 percent discount rate, as well as current year exports (as under the DSF methodology), instead of end-2004 exchange rates and currency-specific CIRR and a three-year backward looking average for exports (as under the HIPC methodology).
3/ The residual captures any accumulation of arrears, the non-delivery of HIPC assistance or changes in the time profile of assistance as well as weaknesses in the underlying bilateral data.

6. To assess the risk for future debt distress, the staffs have developed a baseline scenario and several stress tests that are compared against the indicative thresholds. The debt service numbers underlying the analysis reflect any debt relief still to be delivered after 2004. Sections B and C discuss the underlying assumptions and present the results of the analysis, and section D concludes.

The Baseline Scenario

7. The baseline scenario is developed for 2005–25. The medium-term assumptions for 2005–08 are covered in the Staff Report. The main assumptions for the subsequent years are summarized in Box 2.

Box 2. Baseline Assumptions

The baseline scenario is developed for 2005-2025. The medium-term assumptions for 2005-2008 are covered in the staff report. For the subsequent years, the main assumptions are the following:

- Annual real GDP growth is 6 percent. This is in line with the average growth rate over the past 15 years and also reflects gains from sustained efforts to improve private sector development and reduce government involvement in the economy;
- Average inflation as measured by the implicit deflator of GDP is 2.1 percent, owing to the peg of the currency to the euro;
- The revenue to GDP ratio is projected to rise gradually from 13.9 percent in 2005 to 18 Percent by 2016 and stays constant afterwards, reflecting the gradual convergence of the tax revenue ratio to the 17 percent WAEMU norm owing to rising incomes and improved tax collection;
- The expenditure to GDP ratio reaches 24.6 percent of GDP by 2008 from 21.4 percent in 2005, reflecting efforts of the authorities to make progress towards meeting the Millennium Development Goals (MDGs), and afterwards remains constant as a share of GDP.
- Cotton exports, which in 2004 were almost 70 percent of total exports, are projected to grow at 10 percent annually in real terms during 2009–15 and at 5 percent thereafter. These growth rates are more conservative than the five and ten year historical averages, which were 13 and 17 percent respectively. Growth will come from increasing the cotton growing areas and from productivity gains, notably from using better crop varieties and increasing the use of fertilizers.
- Non-cotton exports are assumed to grow at 5 percent in real terms annually during 2009–15 and at 7.5 percent thereafter. The authorities have identified non-cotton exports as a key sector for increasing economic diversification, and recent reforms in the mining sector and ongoing efforts such as building a fruit and vegetables terminal support this strategy.
- The export projections and the assumption that imports grow on average 7.4 percent annually over 2008–25 (slightly less than nominal GDP growth) lead to a gradual improvement of the current account deficit excluding grants relative to GDP, from 11.7 percent in 2008 to 4.9 percent by 2025.
- Finally, the baseline scenario assumes that the grant share of the total external financing need is 57 percent and 54 percent for 2005 and 2006 respectively, calculated from information provided by the authorities. From 2007 onwards, the grant component is projected to remain at 55 percent, which is more conservative than the ten year historic average (60.4 percent).

8. Under the baseline scenario, all but one of the external debt indicators maintain a stable path for the entire projection period and remain well below their respective thresholds (Table 1a). Only the NPV of debt-to-exports ratio breaches its indicative threshold over the medium term, but improves again in the long term. Between 2005 and 2013, the ratio averages 218 percent before falling below 200 percent in 2014. The medium-term deterioration in the NPV of debt-to-exports ratio is due to several factors, including the medium-term run-up in expenditures (from 21.4 percent of GDP in 2004 to 24.6 percent in 2006).

9. The deterioration in the ratio is compounded by the relatively weak export growth in the short term, which in dollar terms is projected to grow by 5½ percent on average in 2005-06 compared to an average growth of 25 percent in 2002-04. Other debt burden indicators such as the debt service ratios and the NPV of debt-to-GDP and revenue ratios remain stable over the projection period. Until 2025, the NPV of debt-to-GDP ratio is projected to average 23 percent, and Burkina Faso would spend on average 9 percent of its export proceeds or 7 percent of its total revenues on the payment of debt service.

Sensitivity Analysis

10. The sensitivity analysis shows that, again with the exception of the NPV of debt-to-exports ratio, the indicators remain below their respective indicative thresholds, even under stress (Table 1b). The projections for the NPV of debt-to-exports ratio and the debt service ratios are most vulnerable to the following three standard scenarios: (i) should export projections remain subdued and grow at one standard deviation below historical levels, (ii) in the event of a combined shock or (iii) if the terms of financing available to the country worsen (bound tests B2 and B5 and scenario A2 in Table 1b respectively). The staffs performed an additional scenario that assumes that the authorities achieve only half of the projected increase in the revenue-to-GDP target 2015 (Table 1c). Under this scenario, should the authorities not be able to compensate the lower revenues with expenditure cuts or a higher grant share of their financing, the various debt ratios would not improve over the long run, and the NPV of debt-to-exports ratio would remain above 200 percent over the entire projection period. Finally, under the standard historical scenario (scenario A1, Table 1b), all indicators exhibit a sharp upward trend in the outer years of the projection period. The main reason for the worsening of the indicators compared to the baseline is that the historical scenario assumes that the non-interest current account remains at 8.9 percent of GDP leading to a stark increase in new borrowing.[14]

[14] The historical scenario also does not take into account the forward-looking export price projections from the latest WEO.

Staff Assessment

11. The findings from Debt Sustainability Analysis lead the staffs to conclude that while Burkina Faso's debt dynamics are sustainable, the country faces a moderate risk of debt distress. The debt situation is deemed sustainable because, with the exception of the NPV of debt-to-exports ratio, all debt indicators remain well below the policy-contingent indicative thresholds, both under the baseline and under the alternative scenarios. The NPV of debt-to-exports ratio breaches the indicative threshold for some years under the baseline before improving gradually in the outer years of the projection period. Taken as a whole, the staffs do not view the temporary breach of that threshold as indicative of a fundamental debt problem.

12. However, the results from the alternative scenarios underscore the need for the authorities to continue to implement their program of sound macroeconomic policies and reforms, including towards achieving higher export growth and other growth-enhancing measures. The alternative revenue scenario highlights the importance of maintaining efforts to improve revenue collection. Those factors, combined with unwavering resolve to attract financing at favorable terms—preferably through grants—would help Burkina Faso in achieving progress towards the MDGs (as reflected in the ambitious medium-term expenditure program) while containing the risks to external debt sustainability.

Table 1a. Burkina Faso: External Debt Sustainability Framework, Baseline Scenario, 2002-2025 1/
(In percent of GDP, unless otherwise indicated)

	Actual	Historical Average 6/	Standard Deviation 6/	Estimate				Projections					
	2004			2005	2006	2007	2008	2009	2010	2005-10 Average	2015	2025	2011-25 Average
External debt (nominal) 1/	**34.3**			**34.8**	**37.3**	**38.7**	**39.6**	**40.5**	**41.1**	**38.7**	**40.1**	**35.1**	38.4
Change in external debt	-8.6			0.5	2.6	1.4	0.9	0.9	0.6	1.1	-0.8	-0.4	-0.4
Identified net debt-creating flows	0.7			6.8	10.2	8.5	8.0	6.4	5.7	7.6	2.8	2.6	3.2
Non-interest current account deficit	**7.6**	**8.9**	**2.8**	**8.9**	**11.5**	**10.6**	**10.4**	**8.6**	**7.9**	9.6	**4.9**	**4.4**	5.1
Deficit in balance of goods and services	11.1			12.8	13.4	12.5	12.2	11.8	11.3	12.3	7.9	6.7	7.9
Exports	9.9			9.6	9.6	9.8	10.0	10.3	10.6	10.0	13.2	12.8	12.7
Imports	21.0			22.4	23.0	22.2	22.3	22.1	21.9	22.3	21.1	19.5	20.6
Net current transfers (negative = inflow)	-3.9	-5.5	2.2	-4.3	-2.2	-2.1	-2.0	-3.2	-3.4	-2.9	-2.9	-2.1	-2.7
Other current account flows (negative = net inflow)	0.3			0.3	0.3	0.2	0.1	0.1	0.0	0.2	-0.1	-0.2	-0.1
Net FDI (negative = inflow)	**-0.4**	**-0.4**	**0.2**	**-0.4**	**-0.3**	**-0.3**	**-0.3**	**-0.3**	**-0.3**	-0.3	**-0.1**	**-0.1**	-0.1
Endogenous debt dynamics 2/	**-6.5**			**-1.7**	**-1.0**	**-1.8**	**-2.1**	**-1.9**	**-2.0**	-1.7	**-1.9**	**-1.6**	-1.8
Denominator: 1+g+ρ+gρ	1.2			1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1
Contribution from nominal interest rate	0.3			0.3	0.2	0.3	0.3	0.3	0.3	0.3	0.4	0.3	0.4
Contribution from real GDP growth	-5.4			-2.0	-1.2	-2.0	-2.3	-2.2	-2.3	-2.0	-2.3	-2.0	-2.2
Contribution from price and exchange rate changes	-1.4			…	…	…	…	…	…	...	…	…	
Residual (3-4) 3/	**-9.3**			**-6.2**	**-7.6**	**-7.1**	**-7.1**	**-5.6**	**-5.1**	-6.4	**-3.6**	**-3.0**	-3.6
o/w exceptional financing	-0.5			-0.7	-0.5	-0.3	-0.3	-0.3	-0.2	-0.4	-0.1	0.0	-0.1
NPV of external debt 4/	18.1			19.2	20.9	21.9	22.6	23.4	24.0	22.0	24.5	22.2	23.7
In percent of exports	183.1			200.5	216.9	223.6	225.6	226.8	226.0	219.9	185.6	173.1	187.3
In percent of revenues	141.5			143.9	149.3	151.8	153.7	154.9	154.9	151.4	133.8	123.6	129.3
Debt service-to-exports ratio (in percent)	4.9			8.1	9.1	9.9	9.6	8.8	8.7	9.1	8.5	10.3	9.3
Debt service-to-revenues ratio (in percent)	3.8			5.8	6.2	6.8	6.6	6.0	6.0	6.2	6.1	7.3	6.9
Total gross financing need (millions of U.S. dollars)	393.4			512.0	703.8	709.0	759.5	689.5	691.7	677.6	702.7	1,472.4	964.4
Non-interest current account deficit that stabilizes debt ratio	16.2			8.3	8.9	9.2	9.5	7.8	7.3	8.5	5.7	4.8	5.5
Key macroeconomic assumptions													
Real GDP growth (in percent)	15.0	7.3	11.6	6.3	3.7	5.9	6.6	6.1	6.1	5.8	6.0	6.0	6.0
GDP deflator in US dollar terms (change in percent)	3.3	2.2	3.2	1.7	2.2	2.1	2.2	2.1	2.1	2.1	2.1	2.1	2.1
Effective interest rate (percent) 5/	0.7	0.4	0.4	0.8	0.7	0.7	0.8	0.8	0.9	0.8	1.0	1.1	1.0
Growth of exports of G&S (US dollar terms, in percent)	32.5	9.5	21.1	4.6	6.6	9.8	11.7	11.4	11.6	9.3	13.3	8.0	9.6
Growth of imports of G&S (US dollar terms, in percent)	14.9	9.1	15.7	15.2	8.8	4.4	9.1	7.5	7.5	8.7	7.4	7.4	7.4
Grant element of new public sector borrowing (in percent)	...	...	...	44.9	45.4	46.1	46.1	46.3	46.4	45.9	49.0	44.2	46.5
Memorandum item:													
Nominal GDP (millions of US dollars)	5,110			5,525	5,850	6,323	6,888	7,461	8,083		12,001	26,458	

Source: Staff simulations.

1/ Includes public external debt.
2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that NPV of private sector debt is equivalent to its face value.
5/ Current-year interest payments devided by previous period debt stock.
6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 1b. Burkina Faso: Sensitivity Analyses for Key Indicators of Public External Debt, 2005-25
(In percent)

	Estimate	Projections						
	2005	2006	2007	2008	2009	2010	2015	2025
NPV of debt-to-GDP ratio								
Baseline	19.2	20.9	21.9	22.6	23.4	24.0	24.5	22.2
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2006-25 1/	19.2	18.8	18.7	18.6	19.4	20.5	27.2	38.9
A2. New public sector loans on less favorable terms in 2006-25 2/	19.2	22.1	24.3	26.1	27.9	29.4	33.0	34.6
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2006-07	19.2	22.6	26.2	27.2	28.1	28.8	29.4	26.7
B2. Export value growth at historical average minus one standard deviation in 2006-07 3/	19.2	21.8	24.5	25.2	25.8	26.4	26.2	22.8
B3. US dollar GDP deflator at historical average minus one standard deviation in 2006-07	19.2	21.5	23.3	24.1	24.9	25.6	26.0	23.7
B4. Net non-debt creating flows at historical average minus one standard deviation in 2006-07 4/	19.2	20.0	20.1	20.9	21.8	22.5	23.4	21.9
B5. Combination of B1-B4 using one-half standard deviation shocks	19.2	20.7	22.7	23.6	24.5	25.3	26.1	24.2
B6. One-time 30 percent nominal depreciation relative to the baseline in 2006 5/	19.2	29.5	30.9	32.0	33.1	34.0	34.6	31.5
NPV of debt-to-exports ratio								
Baseline	200.5	216.9	223.6	225.6	226.8	226.0	185.6	173.1
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2006-25 1/	200.5	195.1	190.8	185.6	188.5	193.0	206.5	303.0
A2. New public sector loans on less favorable terms in 2006-25 2/	200.5	229.8	248.3	260.4	270.2	276.6	250.7	269.2
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2006-07	200.5	216.9	223.6	225.6	226.8	226.0	185.6	173.1
B2. Export value growth at historical average minus one standard deviation in 2006-07 3/	200.5	272.8	375.5	375.7	375.0	371.4	297.2	266.2
B3. US dollar GDP deflator at historical average minus one standard deviation in 2006-07	200.5	216.9	223.6	225.6	226.8	226.0	185.6	173.1
B4. Net non-debt creating flows at historical average minus one standard deviation in 2006-07 4/	200.5	207.9	205.7	209.0	211.2	211.5	177.2	170.1
B5. Combination of B1-B4 using one-half standard deviation shocks	200.5	223.5	252.8	256.2	258.4	258.4	215.1	204.5
B6. One-time 30 percent nominal depreciation relative to the baseline in 2006 5/	200.5	216.9	223.6	225.6	226.8	226.0	185.6	173.1
Debt service-to-exports ratio								
Baseline	8.1	9.1	9.9	9.6	8.8	8.7	8.5	10.3
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2006-25 1/	8.1	8.8	9.2	8.7	7.7	7.6	7.2	13.8
A2. New public sector loans on less favorable terms in 2006-25 2/	8.1	9.1	10.5	11.0	10.9	11.0	11.8	16.2
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2006-07	8.1	9.1	9.9	9.6	8.8	8.7	8.5	10.3
B2. Export value growth at historical average minus one standard deviation in 2006-07 3/	8.1	10.9	15.2	15.2	13.9	13.7	14.2	16.1
B3. US dollar GDP deflator at historical average minus one standard deviation in 2006-07	8.1	9.1	9.9	9.6	8.8	8.7	8.5	10.3
B4. Net non-debt creating flows at historical average minus one standard deviation in 2006-07 4/	8.1	9.1	9.8	9.3	8.5	8.5	7.9	10.0
B5. Combination of B1-B4 using one-half standard deviation shocks	8.1	9.8	11.7	11.3	10.3	10.2	9.7	12.0
B6. One-time 30 percent nominal depreciation relative to the baseline in 2006 5/	8.1	9.1	9.9	9.6	8.8	8.7	8.5	10.3
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	44.8	44.8	44.8	44.8	44.8	44.8	44.8	44.8

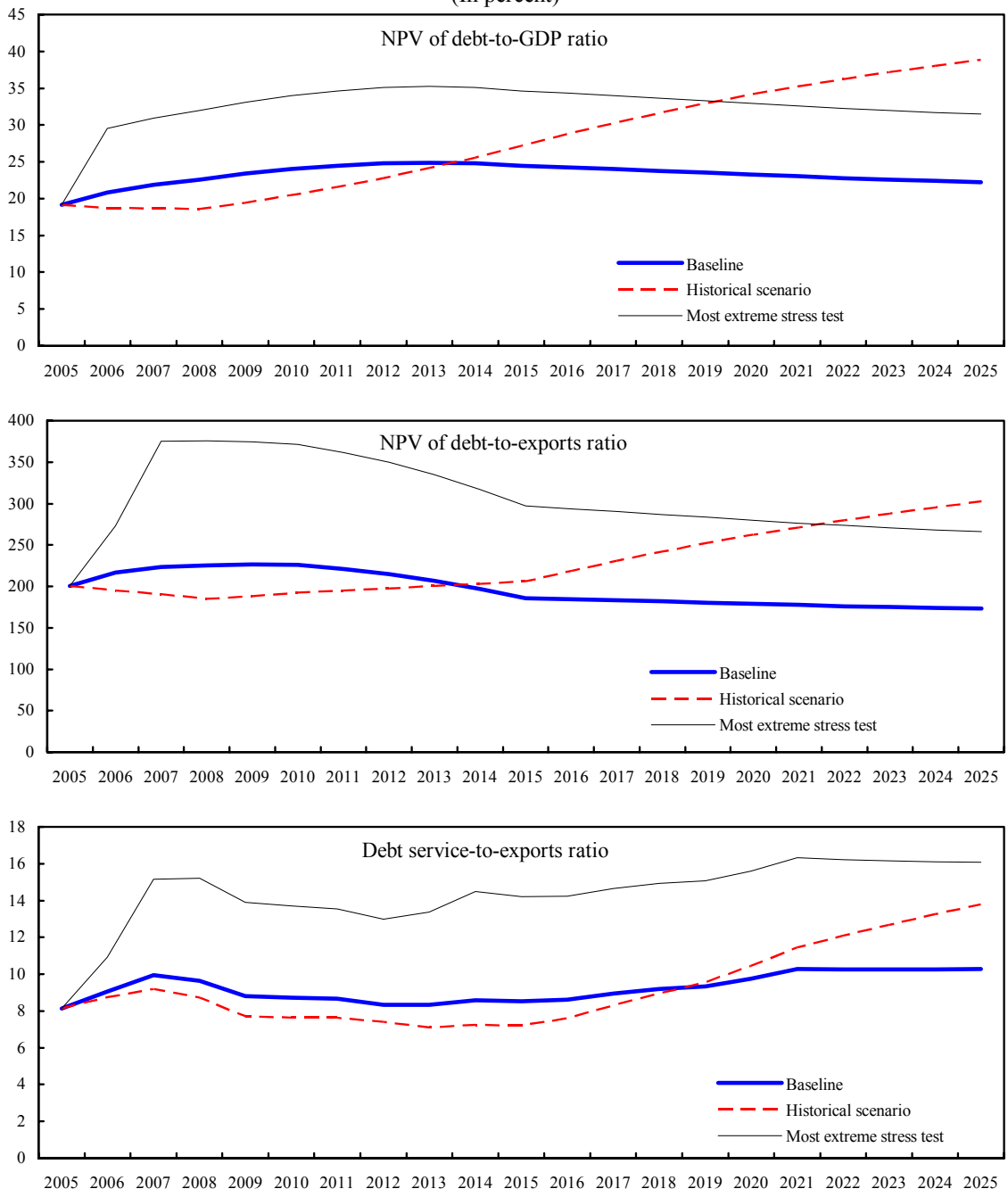
Source: Staff projections and simulations.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 1c. Burkina Faso: Alternative Revenue Scenario for Key Indicators of Public External Debt, 2005-25
(In percent)

	Estimate	Projections						
	2005	2006	2007	2008	2009	2010	2015	2025
Baseline								
Revenue-to-GDP ratio	13.3	14.0	14.4	14.7	15.1	15.5	18.3	18.0
Debt-to-GDP ratio	34.8	37.3	38.7	39.6	40.5	41.1	40.1	35.1
NPV of debt-to-GDP ratio	19.2	20.9	21.9	22.6	23.4	24.0	24.5	22.2
NPV of debt-to-exports ratio	200.5	216.9	223.6	225.6	226.8	226.0	185.6	173.1
NPV of debt-to-revenues ratio	143.9	149.3	151.8	153.7	154.9	154.9	133.8	123.6
Debt service-to-exports ratio	8.1	9.1	9.9	9.6	8.8	8.7	8.5	10.3
Debt service-to-revenues ratio	5.8	6.2	6.8	6.6	6.0	6.0	6.1	7.3
Alternative Scenario: Smaller Revenue Improvement over 2005-2025								
Revenue-to-GDP ratio	13.3	13.6	13.8	14.1	14.3	14.6	15.8	15.8
Debt-to-GDP ratio	34.8	37.8	39.3	40.5	41.7	42.6	44.8	43.9
NPV of debt-to-GDP ratio	19.2	21.1	22.3	23.2	24.2	25.1	27.7	28.7
NPV of debt-to-exports ratio	200.5	219.7	228.1	231.6	234.6	235.9	210.5	223.3
NPV of debt-to-revenues ratio	143.9	155.9	161.4	165.1	169.0	172.1	175.6	181.6
Debt service-to-exports ratio	8.1	9.1	10.0	9.8	9.0	8.9	9.3	13.2
Debt service-to-revenues ratio	5.8	6.4	7.1	7.0	6.5	6.5	7.8	10.7

Figure 1. Burkina Faso: Indicators of Public External Debt
Under Alternative Scenarios, 2005-2025
(In percent)



Source: Staff projections and simulations.